Solids, liquids and gases

Mark scheme and worked solutions for 30 questions: 10 multiple choice, 20 structured.

1 · Part 1

Paper 1 & 2 | Units | [1]

Answer: B

Explanation

Density is mass divided by volume, so its unit is kg/m^3; pressure is measured in pascals (Pa). A gives the unit of force (N) for pressure. C uses kg/m^2, which is mass per area, not per volume. D multiplies instead of dividing and gives the unit of energy for pressure.

2 · Part 1

Paper 1 & 2 | Units | [1]

Answer: D

Worked solution

To convert cm^3 to m^3, divide by 10^6: 450 ÷ 1 000 000 = 0.00045 m^3 = 4.5×10^{-4} m^3.

3 · Part 1

Paper 1 & 2 | Density and pressure | [1]

Answer: A

Worked solution

In SI units: m = 0.540 kg, V = 200×10^{-6} m^3 = 2.0×10^{-4} m^3. ρ = 0.540 ÷ (2.0×10^{-4}) = 2700 kg/m^3.

4 · Part 1

Paper 1 & 2 | Density and pressure | [1]

Answer: C

Explanation

At a point in a liquid at rest, the pressure acts equally in all directions. That is why the sides and the bottom of a container are both pushed on, and why an object under water is pushed on from below as well as from above. A, B and D all wrongly say the pressure has a preferred direction.

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5 · Part 1 Paper 2 only Change of state [1]

Answer: A

Explanation

Evaporation happens only at the surface of a liquid and at any temperature. Boiling (B) happens throughout the liquid and only at the boiling point. Condensation (C) is a gas turning into a liquid, and melting (D) is a solid turning into a liquid.

6 · Part 1 Paper 2 only Change of state [1]

Answer: D

Explanation

Specific heat capacity is the energy required to change the temperature of an object by one degree Celsius per kilogram of mass. A describes melting, not a temperature change. B leaves out "per kilogram". C describes heating through a large temperature range, not 1 °C.

7 · Part 1 Paper 2 only Change of state [1]

Answer: B

Worked solution

Always use the change in temperature: $\Delta T = 60 - 20 = 40$ °C. Then $\Delta Q = 0.50 \times 4200 \times 40 = 84\,000$ J.

8 · Part 1 Paper 1 & 2 Ideal gas molecules [1]

Answer: C

Explanation

Temperature in K = temperature in °C + 273, so −23 + 273 = 250 K. A calculates 23 − 273, which gives a negative Kelvin temperature; that is impossible. B ignores both the sign and the 273. D adds 273 to +23 instead of −23.

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9 · Part 1

Paper 1 & 2 | Ideal gas molecules | [1]

Answer: A

Explanation

When the temperature of a gas rises, its molecules gain kinetic energy, so their average speed increases. B is wrong because molecules in a gas always have a range of speeds. C is wrong because the molecules themselves do not change size. D is wrong because the speed depends on temperature, not on the mass of gas.

10 · Part 1

Paper 1 & 2 | Ideal gas molecules | [1]

Answer: C

Worked solution

p1V1 = p2V2. If V2 = V1 ÷ 2, then p2 = p1V1 ÷ (V1 ÷ 2) = 2p1.

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11 · Part 1 Paper 1 & 2 Density and pressure [7]

Mark scheme

(a) **[1]**

- (top-pan) balance (1)

ALLOW electronic balance OR digital scales

REJECT newtonmeter OR spring balance

(b) **[1]**

- 68.0 − 50.0 = 18.0 cm^3 (1)

(c) **[3]**

- conversion of mass to 0.0576 kg OR volume to 1.80×10^{-5} m^3 (1)
- density = mass / volume seen or used, e.g. $0.0576 \div (1.80 \times 10^{-5})$ OR $57.6 \div 18.0 = 3.2$ g/cm^3 (1)
- 3200 kg/m^3 (1)

ALLOW the conversion mark for converting 3.2 g/cm^3 to kg/m^3 (× 1000) at the end

ECF from (b)

(d) **[2]**

Any two from:

- read the scale at eye level (to avoid parallax) (1)
- read the bottom of the meniscus (1)
- lower the stone gently so that no water splashes out (1)
- make sure the stone is completely under the water (1)
- remove any air bubbles stuck to the stone (1)
- use a measuring cylinder with a smaller scale division OR narrower cylinder (1)

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Worked solution

The stone pushes aside its own volume of water, so its volume is the rise in the water level: 68.0 − 50.0 = 18.0 cm^3. Its density is mass ÷ volume. In g/cm^3: $57.6 \div 18.0 = 3.2$ g/cm^3. To change to kg/m^3, multiply by 1000, giving 3200 kg/m^3. (Check in SI units: $0.0576 \text{ kg} \div (1.80 \times 10^{-5} \text{ m}^3) = 3200$ kg/m^3.) The volume reading is the least precise measurement, so read it carefully at eye level from the bottom of the meniscus, and lower the stone gently so no water is lost.

12 · Part 1 Paper 1 & 2 Density and pressure [8]

Mark scheme

(a) **[1]**

- pressure = force / area OR p = F / A (1)

ALLOW rearranged forms, e.g. force = pressure × area

(b) **[3]**

- area = 1.2 × 0.50 = 0.60 m^2 (1)
- pressure = 1800 ÷ 0.60 (1)
- 3000 Pa (1)

ALLOW 3000 N/m^2

(c) **[2]**

- total area = 4 × 0.0050 = 0.020 m^2 (1)
- pressure = 1800 ÷ 0.020 = 90 000 Pa (1)

ALLOW 9.0×10^4 Pa

ALLOW 1 mark for 360 000 Pa (area of only one foot used)

(d) **[2]**

- the force (weight) on the floor is the same (1)
- but it acts on a much smaller area, so the pressure is (much) greater (1)

Worked solution

The base area is 1.2 m × 0.50 m = 0.60 m^2, so the pressure is 1800 N ÷ 0.60 m^2 = 3000 Pa. On the feet, the same 1800 N is spread over only 4 × 0.0050 = 0.020 m^2, so the pressure is 1800 ÷ 0.020 = 90 000 Pa, thirty times bigger. The weight has not changed; it is the much smaller contact area that makes the pressure large enough to dent a soft floor.

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13 · Part 1 Paper 1 & 2 Density and pressure [6]

Mark scheme

(a) **[1]**

- pressure difference = height × density × gravitational field strength OR p = h × ρ × g (1)

(b) **[2]**

- 25 × 1030 × 10 (1)
- 257 500 Pa (1)

ALLOW 2.6 × 10^5 Pa OR 258 000 Pa

(c) **[1]**

- 257 500 + 100 000 = 357 500 Pa OR 3.6 × 10^5 Pa (1)

ECF from (b)

(d) **[2]**

- the pressure stays the same (1)
- because the depth (height of water above the diver) does not change (1)

ALLOW pressure depends only on depth (density and g are unchanged)

Worked solution

The pressure difference due to the water is p = h × ρ × g = 25 × 1030 × 10 = 257 500 Pa. The air above the sea also pushes down, so the total pressure is 257 500 + 1.0 × 10^5 = 357 500 Pa, about 3.6 × 10^5 Pa. Pressure in a liquid depends on depth, not on horizontal position, so swimming sideways at 25 m leaves the pressure unchanged.

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14 · Part 1 Paper 1 & 2 Density and pressure [7]

Mark scheme

(a) **[2]**

- 8.0 × 1000 × 10 (1)
- 80 000 Pa (1)

ALLOW 8.0×10^4 Pa

(b) **[3]**

- force = pressure × area OR rearrangement of p = F / A (1)
- $80\,000 \times 2.0 \times 10^{-4}$ (1)
- 16 N (1)

ECF from (a)

(c) **[2]**

- pressure difference = height × density × g, which does not include area OR width (1)
- the depth (height), density and g are all unchanged, so the pressure difference is unchanged (1)

Worked solution

The pressure difference is $p = h \times \rho \times g = 8.0 \times 1000 \times 10 = 80\,000$ Pa. Pressure is force per unit area, so the force on the closed tap is $F = p \times A = 80\,000 \times 2.0 \times 10^{-4} = 16$ N. A wider tank holds more water, but the pressure at a point depends only on the depth below the surface, the density and g. None of these has changed, so the pressure at the tap is still 80 000 Pa.

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15 · Part 1 Paper 1 & 2 Density and pressure [7]

Mark scheme

(a) **[2]**

- independent variable: volume of oil (1)
- dependent variable: mass (of cylinder and oil) (1)

(b) **[3]**

- mass of oil found from a difference, e.g. 157.6 − 84.0 = 73.6 g for 80 cm^3 (1)
- density = 73.6 ÷ 80 = 0.92 g/cm^3 (1)
- 920 kg/m^3 (1)

ALLOW any correct pair of readings, or the gradient 18.4 g per 20 cm^3

ALLOW 2 marks for 0.92 g/cm^3 with no conversion

ALLOW 1 mark for 157.6 ÷ 80 = 1.97 (empty cylinder mass not subtracted) leading to 1970 kg/m^3

(c) **[2]**

Any two from:

- an anomalous reading can be identified (1)
- the effect of random errors (e.g. in reading the meniscus) is reduced (1)
- the density can be found from several values OR from a graph gradient and averaged, giving a more reliable result (1)
- it shows the mass increases by the same amount for each 20 cm^3, i.e. a consistent pattern (1)

Worked solution

The student changes the volume of oil, so that is the independent variable; he measures the mass, so that is the dependent variable. The empty cylinder has a mass of 84.0 g, so the oil alone has a mass of 157.6 − 84.0 = 73.6 g when its volume is 80 cm^3. Density = 73.6 ÷ 80 = 0.92 g/cm^3 = 920 kg/m^3. Each extra 20 cm^3 adds 18.4 g, which gives the same answer (18.4 ÷ 20 = 0.92). Taking several readings lets you spot a mistake and reduces the effect of random errors.

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16 · Part 1 Paper 2 only Change of state [6]

Mark scheme

(a) **[2]**

- (particles) close together in a regular pattern OR fixed arrangement (1)
- vibrate about fixed positions (1)

(b) **[2]**

- (particles) far apart, randomly arranged (1)
- move quickly and randomly in all directions (1)

(c) **[2]**

- in a gas the particles are far apart, with lots of space between them (1)
- in a solid the particles are already touching OR very close together, so there is no space to push them into (1)

Worked solution

In a solid the particles are packed closely in a regular pattern and can only vibrate about fixed positions. In a gas the particles are far apart with no regular arrangement, and they move quickly and randomly in all directions. A gas can be squashed because most of its volume is empty space between the particles; pushing the particles closer together is easy. A solid's particles are already as close as they can be, so it cannot be squashed noticeably.

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17 · Part 1 Paper 2 only Change of state [6]

Mark scheme

(a) **[1]**

- 69 °C (1)

(b) **[1]**

- 3 min and 6 min (1)

ALLOW 3 to 6 minutes OR 3–6 min in any form

REJECT 2 and 7 minutes (the temperature is still changing at these times)

(c) **[3]**

- the stearic acid is changing state OR freezing OR solidifying (1)
- forces OR bonds form between the particles as they become fixed in place, releasing energy (1)
- this released energy balances the energy transferred to the surroundings, so the (average) kinetic energy of the particles does not change OR the energy stored decreases without a fall in temperature (1)

(d) **[1]**

Any one from:

- take readings more often (e.g. every 30 s) (1)
- use a thermometer with a finer scale OR a temperature sensor and data logger (1)
- stir the liquid before it solidifies so the temperature is the same throughout (1)

Worked solution

The temperature stops falling at 69 °C and stays there from 3 to 6 minutes; this flat section shows the change of state, so the melting (freezing) point is 69 °C. During this time the liquid is turning into a solid. As the particles move into fixed positions, bonds form between them and energy is released. This energy replaces what is being lost to the air, so the particles' kinetic energy, and therefore the temperature, stays the same until all the stearic acid has solidified. Readings taken more often, or with a more precise thermometer, would pin down the plateau temperature more accurately.

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18 · Part 1 Paper 2 only Change of state [7]

Mark scheme

(a) **[3]**

- change in temperature = 42 − 18 = 24 °C (1)
- 1.5 × 900 × 24 (1)
- 32 400 J (1)

ALLOW 32 000 J OR 3.2×10^4 J

(b) **[3]**

- $\Delta T = \Delta Q \div (m \times c)$ (1)
- 32 400 ÷ (1.5 × 4200) OR 32 400 ÷ 6300 (1)
- 5.1 °C (1)

ALLOW 5.14 °C

ECF from (a)

(c) **[1]**

- water has a (much) higher specific heat capacity, so more energy is needed to raise the temperature of each kilogram by 1 °C (1)

Worked solution

The temperature change of the aluminium is 42 − 18 = 24 °C, so $\Delta Q = 1.5 \times 900 \times 24 = 32\,400$ J. For the water, rearrange: $\Delta T = \Delta Q \div (m \times c) = 32\,400 \div (1.5 \times 4200) = 32\,400 \div 6300 = 5.1$ °C. Check: the masses are equal, so the temperature rises are in the inverse ratio of the specific heat capacities: 24 × 900 ÷ 4200 = 5.1 °C. Water needs 4200 J to warm each kilogram by 1 °C, compared with only 900 J for aluminium, so the same energy warms water much less.

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19 · Part 1 Paper 2 only Change of state [7]

Mark scheme

(a) **[3]**

- change in temperature = 31.5 − 19.0 = 12.5 °C (1)
- c = 6000 ÷ (1.00 × 12.5) (1)
- 480 J/kg °C (1)
- final answer mark requires the correct unit

(b) **[2]**

- some of the energy supplied is transferred to the surroundings (OR to the heater and thermometer) instead of the block (1)
- so the temperature rise is smaller than it would be, making the calculated value too large OR the 6000 J used in the calculation is more than the energy that actually heated the block (1)

(c) **[2]**

Any two from:

- wrap the block in insulation OR place it on an insulating mat (1)
- put a few drops of oil OR water in the thermometer hole for good thermal contact (1)
- put oil OR water in the heater hole for good thermal contact (1)
- use a larger temperature rise OR read the highest temperature after the heater is switched off (1)

ALLOW use a lid OR cover (reduce energy transfer to the air)

Worked solution

The temperature rise is 31.5 − 19.0 = 12.5 °C. Rearranging $\Delta Q = m \times c \times \Delta T$ gives $c = \Delta Q \div (m \times \Delta T) = 6000 \div (1.00 \times 12.5) = 480$ J/kg °C. Check: $1.00 \times 480 \times 12.5 = 6000$ J. The value is too high because not all 6000 J went into the block: some escaped to the air and the bench. The block therefore warmed less than it would have, and a smaller ΔT in the denominator makes c too large. Insulating the block and filling the holes with oil (so the heater and thermometer are in good thermal contact) reduce this error.

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20 · Part 1 Paper 2 only Change of state [6]

Mark scheme

(a) **[1]**

- liquid to gas OR water turns into steam OR water vapour (1)

(b) **[5]**

Indicative points, any five from:

- heating transfers energy to the water, so the energy stored in the water increases (throughout) (1)
- from 20 °C to 100 °C the (average) kinetic energy of the particles increases (1)
- the particles move faster, so the temperature rises (1)
- at 100 °C the energy supplied is used to change state (liquid to gas), not to raise the temperature (1)
- the energy is used to break the bonds OR overcome the forces between the particles so they can separate (1)
- the (average) kinetic energy of the particles does not increase, so the temperature stays at 100 °C while the water boils (1)
- the particles in the gas are much further apart and move freely OR randomly (1)

Worked solution

Throughout the heating, energy is transferred to the water, so the energy stored in it keeps increasing. At first this energy makes the particles move faster: their average kinetic energy rises, and so the temperature rises from 20 °C to 100 °C. At 100 °C the water boils. Now the energy supplied is used to break the bonds between the particles so that they can separate and escape as a gas, far apart and moving freely. Because the energy is not increasing the particles' kinetic energy, the temperature stays at 100 °C until all the water has boiled away. Heating therefore either raises the temperature or produces a change of state.

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21 · Part 1 Paper 2 only Change of state [7]

Mark scheme

(a) **[3]**

- change in temperature = 65 − 25 = 40 °C (1)
- 120 × 880 × 40 (1)
- 4 224 000 J (1)

ALLOW 4.2 × 10^6 J OR 4.2 MJ

(b) **[2]**

- mass = 4 224 000 ÷ (4200 × 40) (1)
- 25 kg (1)

ALLOW 25.1 kg

ECF from (a)

(c) **[2]**

- a material with a high specific heat capacity stores a lot of energy for each kilogram and each degree of temperature change (1)
- so a smaller mass is needed OR more energy is released into the room as it cools through a given temperature range (1)

Worked solution

The concrete cools by 65 − 25 = 40 °C, so ΔQ = 120 × 880 × 40 = 4 224 000 J, about 4.2 MJ. For water to release the same energy over the same 40 °C fall, m = ΔQ ÷ (c × ΔT) = 4 224 000 ÷ (4200 × 40) = 25 kg. Check: the mass needed is inversely proportional to c, so 120 × 880 ÷ 4200 = 25 kg. A high specific heat capacity means each kilogram stores a lot of energy for each degree it cools, so a heater can store more energy in a given mass.

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22 · Part 1 Paper 2 only Change of state [5]

Mark scheme

(a) **[2]**

- the energy increases the (average) kinetic energy of the particles OR makes the particles vibrate faster OR more (1)
- temperature is a measure of the (average) kinetic energy of the particles, so the temperature rises (1)

(b) **[3]**

- particles start in a regular pattern and vibrate about fixed positions (1)
- the bonds OR forces between particles are weakened OR partly broken (1)
- in the liquid the particles are still close together but randomly arranged and can move past each other (1)

Worked solution

Below 0 °C, energy transferred to the ice makes its particles vibrate faster about their fixed positions. Their average kinetic energy increases, which is what a rise in temperature means. At 0 °C the ice melts: the particles break free of their fixed positions in the regular pattern. In the water they are still close together, but they are arranged randomly and can slide past each other.

23 · Part 1 Paper 1 & 2 Ideal gas molecules [5]

Mark scheme

(a) **[3]**

- the molecules are moving (quickly and) randomly (1)
- they collide with OR hit the inside surface of the balloon (1)
- each collision exerts a force on the surface, and force per unit area is pressure (1)

(b) **[2]**

- there are more molecules (in the same volume) (1)
- so there are more collisions with the surface each second, giving a larger force (on each unit area) (1)

Worked solution

The air molecules move randomly at high speed. They keep hitting the inside of the balloon, and each hit pushes on the rubber with a small force. All these forces acting on the area of the balloon produce a pressure. In the second balloon there are more molecules in the same volume at the same temperature, so the molecules hit the inside surface more often. A larger total force on the same area means a greater pressure.

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24 · Part 1 Paper 1 & 2 Ideal gas molecules [5]

Mark scheme

(a) **[1]**

- −273 °C (1)

(b) **[2]**

- as a gas is cooled its molecules lose kinetic energy OR slow down (1)
- at absolute zero the molecules have the lowest possible (zero) kinetic energy, so the temperature cannot fall any further (1)

> **ALLOW** the pressure of a gas would be zero at absolute zero

(c) **[1]**

- 77 − 273 = −196 °C (1)

(d) **[1]**

- 20 + 273 = 293 K (1)

Worked solution

Temperature is linked to how fast the particles move. As a gas cools, its molecules slow down and their kinetic energy falls. At −273 °C they would have the least possible kinetic energy, so nothing can be colder: this is absolute zero, 0 K. To convert between the scales, use T (K) = θ (°C) + 273. So 77 K is 77 − 273 = −196 °C, and 20 °C is 20 + 273 = 293 K.

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25 · Part 1 Paper 1 & 2 Ideal gas molecules [7]

Mark scheme

(a) **[2]**

- 27 °C = 300 K (1)
- 177 °C = 450 K (1)

(b) **[2]**

- p2 = 2.0 × 10^5 × 450 ÷ 300 (1)
- 3.0 × 10^5 Pa (1)

ECF from (a)

ALLOW 1 mark for 1.3 × 10^6 Pa (Celsius temperatures used)

(c) **[1]**

- the can is rigid OR made of metal, so it cannot expand (noticeably) (1)

(d) **[2]**

- the pressure of the gas increases as its temperature rises (at constant volume) (1)
- the pressure could become greater than the can can withstand, so the can could burst OR explode (1)

Worked solution

Gas laws must use Kelvin temperatures: 27 + 273 = 300 K and 177 + 273 = 450 K. From p1 / T1 = p2 / T2, p2 = p1 × T2 ÷ T1 = 2.0 × 10^5 × 450 ÷ 300 = 3.0 × 10^5 Pa. Check: the Kelvin temperature has increased by a factor of 1.5, so the pressure increases by the same factor. The metal can does not stretch, so the volume stays fixed. If the can gets too hot, the pressure inside could become greater than the can can withstand, and it could burst.

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26 · Part 1 Paper 1 & 2 Ideal gas molecules [6]

Mark scheme

(a) **[1]**

- so that the temperature of the air stays constant OR the air does not warm up, because p1V1 = p2V2 applies only at constant temperature (1)

(b) **[3]**

- p2 = p1V1 ÷ V2 (1)
- $1.0 \times 10^5 \times 60 \div 24$ (1)
- 2.5×10^5 Pa (1)

(c) **[2]**

- $V = 1.0 \times 10^5 \times 60 \div (0.80 \times 10^5)$ (1)
- 75 cm^3 (1)

ALLOW using 2.5×10^5 Pa and 24 cm^3 as the starting values

Worked solution

The rule p1V1 = p2V2 only works if the temperature stays the same; pushing slowly gives the air time to stay at room temperature. Rearranging, p2 = p1V1 ÷ V2 = $1.0 \times 10^5 \times 60 \div 24 = 2.5 \times 10^5$ Pa. Check: $2.5 \times 10^5 \times 24 = 6.0 \times 10^6$, the same as $1.0 \times 10^5 \times 60$. When the pressure falls to 0.80×10^5 Pa, $V = 6.0 \times 10^6 \div (0.80 \times 10^5) = 75$ cm^3. The volume can be left in cm^3 because the same unit is used on both sides.

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27 · Part 1 Paper 1 & 2 Ideal gas molecules [7]

Mark scheme

(a) **[4]**

Indicative points, any four from:

- the molecules move randomly and collide with the walls (of the cylinder and piston) (1)
- the temperature is constant, so the (average) speed OR kinetic energy of the molecules is unchanged (1)
- each collision exerts the same (average) force as before (1)
- in a smaller volume the molecules are closer together OR there are more molecules per unit volume (1)
- so they hit the walls more often OR more collisions per second (on each unit area) (1)
- a larger force on each unit area means a greater pressure (1)

(b) **[3]**

- heating increases the (average) kinetic energy OR speed of the molecules (1)
- the molecules hit the walls more often (1)
- and with more force in each collision, so the force on each unit area (pressure) increases (1)

Worked solution

Gas pressure comes from molecules hitting the walls. In (a) the temperature does not change, so the molecules move at the same average speed and each collision gives the same force. But the same molecules are now packed into a smaller space, so they reach the walls more often. More collisions each second on each square metre of wall means a larger force per unit area: a higher pressure. In (b) the volume is fixed but the molecules gain kinetic energy and move faster. They hit the walls more often and harder, so again the force per unit area, and therefore the pressure, increases.

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28 · Part 1 Paper 1 & 2 Ideal gas molecules [7]

Mark scheme

(a) **[1]**

- volume of the air OR mass (amount) of air in the flask (1)

(b) **[3]**

- 0 °C = 273 K and 80 °C = 353 K (1)
- p ÷ T: 100 ÷ 273 = 0.366 and 129 ÷ 353 = 0.365 (kPa/K) (1)
- the values are (almost) the same, so p is proportional to T (1)

ALLOW ratio of temperatures 353 ÷ 273 = 1.29 and ratio of pressures 129 ÷ 100 = 1.29, so equal

(c) **[2]**

- p2 = 100 × 373 ÷ 273 (1)
- 137 kPa (1)

ALLOW 136 kPa to 137 kPa (from any pair of readings)

ECF from (b) for Kelvin conversion

(d) **[1]**

- so the air in the flask reaches the same temperature as the water (1)

Worked solution

Only the temperature should change, so the volume and amount of air must stay constant (the flask is sealed). Convert to kelvin: 0 °C = 273 K and 80 °C = 353 K. Then p ÷ T = 100 ÷ 273 = 0.366 and 129 ÷ 353 = 0.365. These are equal within the reading precision, so pressure is proportional to Kelvin temperature. At 100 °C (373 K): p2 = p1 × T2 ÷ T1 = 100 × 373 ÷ 273 = 137 kPa. Waiting lets energy transfer from the water to the air so that the thermometer reading really is the temperature of the gas.

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29 · Part 1 Paper 1 & 2 Ideal gas molecules [6]

Mark scheme

(a) **[1]**

- the Kelvin temperature is (directly) proportional to the average kinetic energy of the molecules (1)

(b) **[3]**

- 27 °C = 300 K (1)
- 327 °C = 600 K (1)
- factor = 600 ÷ 300 = 2 (1)

ALLOW 1 mark only for 327 ÷ 27 = 12 (Celsius used)

(c) **[2]**

- the average kinetic energy is proportional to Kelvin temperature, not Celsius temperature (1)
- 20 °C = 293 K and 40 °C = 313 K, so the Kelvin temperature only increases by a factor of about 1.07 (so the average kinetic energy increases by about 7%) (1)

Worked solution

The average kinetic energy of gas molecules is proportional to the Kelvin temperature. In (b), 27 °C = 300 K and 327 °C = 600 K, so the Kelvin temperature doubles and the average kinetic energy doubles: a factor of 2. In (c), the Celsius numbers double, but the Celsius scale does not start at absolute zero. In kelvin the temperature goes from 293 K to 313 K, a factor of 313 ÷ 293 = 1.07. So the average kinetic energy rises by only about 7%, not 100%.

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30 · Part 1 Paper 1 & 2 Ideal gas molecules [6]

Mark scheme

(a) **[2]**

- 15 × 1000 × 10 (1)
- 150 000 Pa (1)

ALLOW 1.5 × 10^5 Pa

(b) **[1]**

- 150 000 + 100 000 = 2.5 × 10^5 Pa (1)

ECF from (a)

(c) **[2]**

- V2 = 2.5 × 10^5 × 2.0 ÷ (1.0 × 10^5) (1)
- 5.0 cm^3 (1)

ECF from (b)

ALLOW 1 mark for 3.0 cm^3 (water pressure only, atmospheric pressure not added)

(d) **[1]**

- the equation p1V1 = p2V2 only applies when the temperature is constant (1)

ALLOW a temperature change would also change the volume

Worked solution

The water above the bubble causes a pressure difference p = h × ρ × g = 15 × 1000 × 10 = 150 000 Pa. The atmosphere also presses on the surface of the lake, so the total pressure at the bottom is 150 000 + 100 000 = 2.5 × 10^5 Pa. At the surface the pressure is just atmospheric, 1.0 × 10^5 Pa. Using p1V1 = p2V2: V2 = 2.5 × 10^5 × 2.0 ÷ (1.0 × 10^5) = 5.0 cm^3. Check: the pressure falls by a factor of 2.5, so the volume grows by a factor of 2.5. This rule only holds if the temperature stays the same, which is why the water temperature must be uniform.

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